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Combinatorics Notes

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Inhoudsopgave

1 Fundamentals of Combinatorics

1.1 Permutations

1.2 Pigeonhole

The pigeonhole principle

1.3 Bijection

1.4 Generating Functions

1.5 Recursion

Recursion is

Problem

How many ways are there to tile a $2 \times n$ grid o

1.6 Group Theory

Definition

Group A group is a set consisting of an identity element g, a binary operator $\oplus$ which maps two elements to one, and requires every element to have an inverse.

Definition

Fixed Point

Definition

Stabilizer

Definition

Orbit

Theorem

Burnside's Lemma Given a group G and an operator d,

$$|G| = \frac{1}{|X/G|} \sum_{x \in G} |X^d|$$

where $|X^d|$ is the number of fixed points in G, X/G is the number of orbits created by d, and $|G|$ is the number of elements in the set.

Bewijs. From the orbit stabilizer theorem

This theorem is particularly useful in computational problems, where

Problem

Classic How many ways are there to color the edges of a square red and blue given that rotations are considered the same?

Problem

Classic How many ways are there to color the edges of a tetrahedron red and blue given that rotations are considered the same?

Problem

Own How many ways are there to color the vertices of a 100-gon 50 colors given that rotations are considered the same?

2 Advanced Topics

2.1 Reduction

Problem

2.2 Induction

Induction and reduction are two of the more common techniques for combinatorics problems. This technique is often coupled with

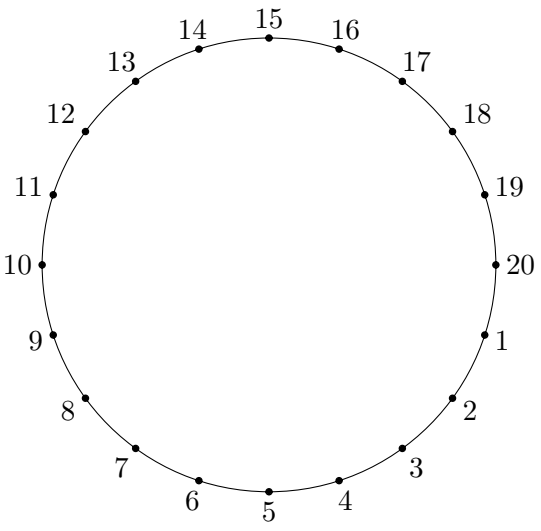
2.3 Invariants and Monovariants

Invariants and monovariants are powerful tools for determining solution sets to problems. Many invariants come in a modulus form or a fixed sum, and many monovariants are used to produce lower bounds to problems.

Problem

2024 ISL C3 Let n be a positive integer. There are $2n$ knights sitting at a round table. They consist of n pairs of partners, each pair of which wishes to shake hands. A pair can shake hands only when next to each other. Every minute, one pair of adjacent knights swaps places. Find the minimum number of exchanges of adjacent knights such that, regardless of the initial arrangement, every knight can meet her partner and shake hands at some time.

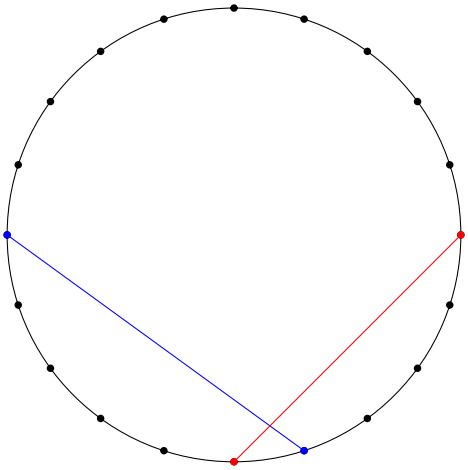
We may quickly determine the starting arrangement which maximizes the number of exchanges: just put every knight diametrically opposite their partner. In the picture below, knights with labels (mod 10) are paired.

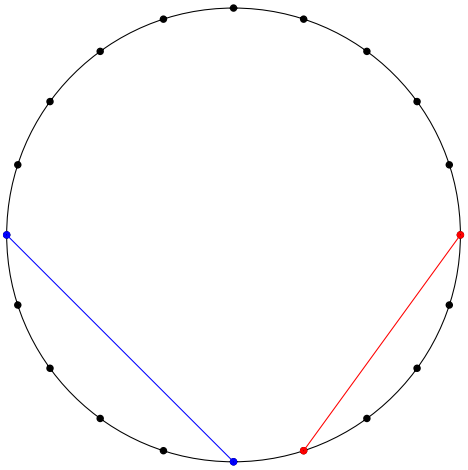


The trouble is that there might be a configuration which is more optimal than this one. Let's break this down.

Intuitively, every move, two knights should move closer to their partners. Given we want to find the maximal number of moves, it suffices to find some variant X where we can simply divide X by 2 to get the answer.

Suppose two pairs of knights are on the same half arc. If their initial position is $ABAB$, then swapping the two in the middle will yield $AABB$, which is desireable. Otherwise, if their position is $ABBA$, swapping the middle ones will be undesireable.



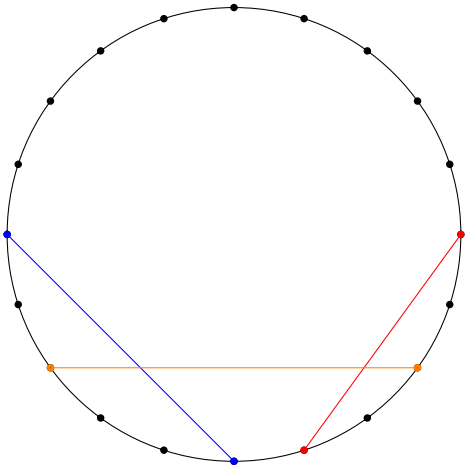


In sum, it is not enough to sum the distances between pairs of knights. We must create the variant

$$F(p) = \sum_{\text{pairs of knights}} \text{distance} - \#\text{ of intersections between chords}$$

Whenever we switch two adjacent knights, the variant decreases by at most 1. But in any position, a move decreasing the distance between a pair of knight decreases the variant, so we have our bound. Then it suffices to maximize F.

Consider what happens when we add a pair of knights to the arrangement. If we add the orange pair, as shown below, we increase the variant by the distance between the two segments. (ok i got lazy and there are an equal number of nodes, but there are supposed to be 22!)



Then the second bound becomes quite obvious: by putting every new pair of knights diametrically opposite, we maximize the sum, which turns out to be $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$, and this is our answer.

This example was very motivated. I encourage you to try thinking of the solution on your own and build your own solution to the problem.

Problem

2025 USAMO 6

Problem

2.4 Graph Theory

Definition

Graph A graph is a collection of vertices (nodes) and edges such

Many combinatorics problems actually have hidden graph interpretations. While apparently simple, graphs can greatly simplify problems and allow us to make concrete arguments about the problem.

Definition

Bipartite A graph is bipartite if it can be divided into two sets A and B such that no two elements in A share an edge, and no two elements in B share an edge.

Visually, a bipartite graph looks like

Theorem

Hall's Marriage Theorem Given an injective function $f : A \to B$, there exists a maximal matching if and only if

$$\forall G \subseteq A, |G| \leq |\bigcup_{S \in \mathcal{F}} S|$$

where $\mathcal{F}$ is the image of the elements of G under f.

Visually, this theorem states that a maximal matching exists if

Problem

2022 ISL C5 Let $m, n \geqslant 2$ be integers, let X be a set with n elements, and let $X_1, X_2, \ldots, X_m$ be pairwise distinct non-empty, not necessary disjoint subset of X. A function $f\colon X \to \{1, 2, \ldots, n+1\}$ is called nice if there exists an index k such that

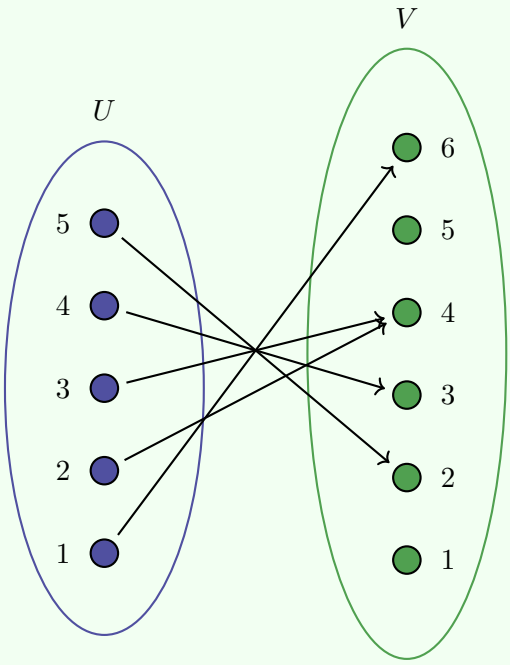
$$\sum_{x \in X_k} f(x) > \sum_{x \in X_i} f(x) \quad \text{for all } i \neq k.$$

Prove that the number of nice functions is at least n^n.

Solution

This problem has a lot of scary notation that we would like to simply. First, we will simplify $F(k) = \sum_{x \in X_k} f(x)$ to be the sum of the elements of in X_k after applying f. Additionally, the strict inequality may be rephrased as: There exists a minimal $F(k)$ over all k.

Now we will create a bipartite graph with n elements on the left side and $n+1$ elements on the right side. Below is a sample graph.



We have the freedom to choose how f maps the elements, and we also know that for any function f, there exists a minimal set, though there does not have to exists a **unique** minimal set.

Solution

Choose a function f such that $f : \{1, 2, 3, \ldots, n\} \to \{2, 3, \ldots, n+1\}$. Then there are n^n such functions satisfying this property.

Without loss of generality, let $X_1, X_2, \ldots, X_k$ be the sets with minimal value. Because the sets are pairwise distinct, there exists an element such that

2.5 Inequalities

Problem

1993 ISL C2 Let $n, k \in \mathbb{Z}^+$ with $k \leq n$ and let S be a set containing n distinct real numbers. Let T be a set of all real numbers of the form $x_1 + x_2 + \ldots + x_k$ where $x_1, x_2, \ldots, x_k$ are distinct elements of S. Prove that T contains at least $k(n-k)+1$ distinct elements.

Solution

Let

$$x_1 < x_2 < \ldots < x_n$$

so that the largest element in T is $x_n + \ldots + x_{n-k+1}$, and the smallest element is $x_1 + \ldots + x_k$. For every sum $x_{i_1} + x_{i_2} + \cdots + x_{i_k)}$, consider the $(n + (n-1) + \ldots + (n-k+1)) - \frac{k(k+1)}{2} + 1 = k(n-k) + 1$ values of $i_1 + \ldots + i_k$. We claim that there is one distinct sum in each set. Indeed, for every element in the set with value m, there is at least one element in the set with value $m+1$ which is greater than it. Then just take the maximal value of the set with value m to finish.

Problem

2013 ISl C4 Let n be a positive integer, and let A be a subset of $\{1, \cdots, n\}$. An A-partition of n into k parts is a representation of n as a sum $n = a_1 + \cdots + a_k$, where the parts $a_1, \cdots, a_k$ belong to A and are not necessarily distinct. The number of different parts in such a partition is the number of (distinct) elements in the set $\{a_1, a_2, \cdots, a_k\}$. We say that an A-partition of n into k parts is optimal if there is no A-partition of n into r parts with $r < k$. Prove that any optimal A-partition of n contains at most $\sqrt[3]{6n}$ different parts.

Solution

At first glance, this problem looks intimidating. *Where* did the $\sqrt[3]{6n}$ come from?

Actually, the bound is very loose; the true asymptote is closer to log in polynomial time, and we have the power to make very weak bounds.

If we take some k element subset $A = \{a_1, a_2, \ldots, a_k\}$ where

$$a_1, a_2, \ldots, a_k \in \{\}$$

Now we're ready to apply some of the

Problem

2023 IMO/5 Let n be a positive integer. A Japanese triangle consists of $1 + 2 + \cdots + n$ circles arranged in an equilateral triangular shape such that for each $i = 1, 2, \ldots, n$, the ith row contains exactly i circles, exactly one of which is coloured red. A ninja path in a Japanese triangle is a sequence of n circles obtained by starting in the top row, then repeatedly going from a circle to one of the two circles immediately below it and finishing in the bottom row. Here is an example of a Japanese triangle with $n = 6$, along with a ninja path in that triangle containing two red circles. [asy] // credit to vEnhance for the diagram (which was better than my original asy): size(4cm); pair X = dir(240); pair Y = dir(0); path c = scale(0.5)*unitcircle; int[] t = 0,0,2,2,3,0; for (int i=0; i¡=5; ++i) for (int j=0; j¡=i; ++j) filldraw(shift(i*X+j*Y)*c, (t[i]==j) ? lightred : white); draw(shift(i*X+j*Y)*c); draw((0,0)–(X+Y)–(2*X+Y)–(3*X+2*Y)–(4*X+2*Y)–(5*X+2*Y),linewidth(1.5)); path q = (3,-3sqrt(3))–(-3,-3sqrt(3)); draw(q,Arrows(TeXHead, 1)); label("$n = 6$", q, S); label("$n = 6$", q, S); [/asy] In terms of n, find the greatest k such that in each Japanese triangle there is a ninja path containing at least k red circles.

Problem

Buffed 2023 IMO P5 Let n be a positive integer. A Japanese triangle consists of $1 + 2 + \cdots + n$ circles arranged in an equilateral triangular shape such that for each $i = 1, 2, \ldots, n$, the ith row contains exactly i circles, exactly one of which is coloured red. A ninja path in a Japanese triangle is a sequence of n circles obtained by starting in the top row, then repeatedly going from a circle to one of the two circles immediately below it and finishing in the bottom row. Here is an example of a Japanese triangle with $n = 6$, along with a ninja path in that triangle containing two red circles.

Problem

2013 ISL C4

Solution

At first glance, this problem looks very scary. *Where* does the $\sqrt[3]{6n}$ even come from?

Actually, this bound is very weak. The asymptote should be closer to $c \log n$ for some constant c. This is because given a sufficiently large set A, we effectively halve n every iteration, without regard to modulus problems.

Now we can calm down. Noting that the $\sqrt[3]{6n}$ looks oddly like the sum of squares formula, we can be confident in making looser bounds.

We may note that a_k is counted at most a_{k+1} times. This is because we can otherwise count a_{k+1} a_k times, meaning there is a partition with less elements.

$$\sum_{i=1}^{k-1} a_i^2 \le \sum_{i=1}^{k-1} a_i a_{i+1} \le n$$

2.6 Constructions

Construction problems can be tricky, especially if the obvious solution seems like a trap. But most constructions are very motivated, and we will give examples of motivated and unmotivated solutions.

Problem

2024 ISL C4 Turbo the snail plays a game on a board with 2024 rows and 2023 columns. There are hidden monsters in 2022 of the cells. Initially, Turbo does not know where any of the monsters are, but he knows that there is exactly one monster in each row except the first row and the last row, and that each column contains at most one monster.

Turbo makes a series of attempts to go from the first row to the last row. On each attempt, he chooses to start on any cell in the first row, then repeatedly moves to an adjacent cell sharing a common side. (He is allowed to return to a previously visited cell.) If he reaches a cell with a monster, his attempt ends and he is transported back to the first row to start a new attempt. The monsters do not move, and Turbo remembers whether or not each cell he has visited contains a monster. If he reaches any cell in the last row, his attempt ends and the game is over.

Problem

Determine the minimum value of n for which Turbo has a strategy that guarantees reaching the last row on the n-th attempt or earlier, regardless of the locations of the monsters.

Here it is. The infamous Turbo the Snail. At first glance, 2023 seems like a fair answer: we just send Turbo down each lane until he wins. But we should find a way to do better.

We should ask what information each death gives us. If Turbo encounters a monster in a column, he knows there are no monsters anywhere else in the column (i.e. anywhere behind it). Then given the encountered monster is not in the first or last column, we require 3 moves to move past it.

Additionally, if we encounter a monster in the $i+1$th row, we need to be sure that there are no monsters in the ith row that would stop us from sneaking past it.

Then we can assume Turbo can sweep out a triangle with base 2021 and height 1011 without encountering a monster. Now we are seemingly stuck. We can't progress without taking a risk, but it's calculated.

Send Turbo all the way to the right. If he encounters a monster, then we are guaranteed a safe column. Otherwise, have him fill in the entire 2023×1011 grid. Otherwise, he hits a monster, and on his second life, have him repeat the same triangle-filling pattern as his first life.

He is guaranteed to hit a monster by pigeonhole, and when he does, he may use the row (which is safe) and then the column he discovered (which is safe) to reach the end.

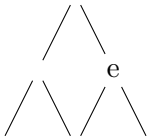
We need at least 3 moves, otherwise we can spawn the monsters on Turbo.

Problem

2022 ISL C6 Let n be a positive integer. We start with n piles of pebbles, each initially containing a single pebble. One can perform moves of the following form: choose two piles, take an equal number of pebbles from each pile and form a new pile out of these pebbles. Find (in terms of n) the smallest number of nonempty piles that one can obtain by performing a finite sequence of moves of this form.

Solution

This problem appears difficult. We may notice that if n is a power of 2, then we can pair all the stones to create a single pile with n stones. Otherwise, we may write n as a binary number and create a pile for each 1 in its binary representation. Below is an example.



Now let us see if we can make any improvements.

Consider the largest bit of n. Because $2^k > 2^{k-1} + \ldots + 1$, we can partition the bit into these smaller sums, and we can select which ones to add to the smallest bit. This means we can reduce the number of piles to 2 which is very good!

Now to show we cannot reduce the number of piles to 1: because every newly-formed pile must be even, we can continually divide whatever numbers we have in 2 to revert to the previous positions.

In the end, we have a bunch of odd numbers, which are impossible to form unless it is 1. Therefore, we cannot get 1 pile for n that is not a power of 2.

This gives us the answer

$$f(n) = \begin{cases} 1 \text{ if } n \text{ is a power of } 2 \\ 2 \text{ otherwise} \end{cases}$$

2.7 Games

2.8 Sledgehammers

2.8.1 Zeckendorf Representation

2.8.2 Wythoff Array

2.8.3

2.9 Algorithms

Problem

2014 ISL C5 A set of lines in the plane is in general position if no two are parallel and no three pass through the same point. A set of lines in general position cuts the plane into regions, some of which have finite area; we call these its finite regions. Prove that for all sufficiently large n, in any set of n lines in general position it is possible to colour at least $\sqrt{n}$ lines blue in such a way that none of its finite regions has a completely blue boundary.

Note: Results with $\sqrt{n}$ replaced by $c\sqrt{n}$ will be awarded points depending on the value of the constant c.

Solution

This problem is tricky.

3 About the Author

4 References and Further Reading